

Harnil Makwana

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WORK EXPERIENCE

Middlesex University

London, UK

Student Learning Assistant (SLA)

Jan 2025 - May 2025

- Guided students in networking labs using Cisco Packet Tracer, emphasizing subnetting, network configuration, and troubleshooting with a focus on secure practices.
- Supported 100+ students across 12+ lab sessions
- Assisted in Python labs by explaining programming concepts, debugging code, and reinforcing secure coding practices.
- Collaborated with faculty to enhance lab exercises by integrating real-world security scenarios and threat mitigation techniques.

Network Lab Maintenance Assistant

Sep 2025 - Present

- Performed regular maintenance on Cisco 2960/3650 switches, ASA 5560 firewalls, and ISR 1941/4221 routers, ensuring stable and secure lab operations.
- Upgraded IOS and ASA firmware using TFTP, USB, and Console, reducing device boot issues and improving system reliability.
- Conducted routine health checks, monitoring interfaces, VLANs, CPU load, and configuration integrity across all devices.

Pantech Instruments

Vadodara

Cloud / Network Support Engineer

May 2021 - Jun 2023

- Engineered and optimized network infrastructure for 50+ devices, reducing downtime by 30% and improving system performance by 20% with proactive threat detection.
- Troubleshoot and resolved 100+ software and hardware issues across Linux and Windows environments, ensuring 99.9% system uptime.
- Developed and maintained secure AWS-based network solutions, reducing operational costs by 15% and reinforcing cloud security best practices.
- Integrated automation tools and advanced threat detection mechanisms resulted in a 40% improvement in system responsiveness.

PROJECT EXPERIENCE

Middlesex University

London, UK

Wireless Penetration Testing of WPA2 Networks

Sep 2025 - Dec 2025

- Led a **4-member team** conducting authorised wireless security testing in a controlled lab environment
- Captured **multiple WPA handshakes** using monitor mode and airodump-ng
- Demonstrated offline password cracking using Aircrack-ng and Hashcat, validating weak passphrase risks
- Evaluated and tested **4 defensive mechanisms** including 802.11w, WPA3-SAE, WPS disabling, and wireless IDS
- Produced structured risk analysis aligned with CIA triad principles

Secure Edge-to-Cloud IoT Architecture

Jan 2025 - Apr 2026

- Led system architecture and AWS security design for an IoT solution connecting **5+ edge sensor nodes** to cloud infrastructure
- Implemented secure MQTT communication with mutual TLS between Raspberry Pi devices and AWS IoT Core
- Configured OpenVPN on EC2 to establish encrypted site-to-cloud connectivity
- Designed a serverless processing pipeline using AWS Lambda and DynamoDB for real-time data handling
- Deployed containerised services via Docker and Kubernetes (EKS) for scalable workloads
- Built a web dashboard enabling authorised access to **real-time and historical IoT sensor data streams**
- Integrated CloudWatch and GuardDuty for continuous monitoring and log analysis

Advanced Hydropower Monitoring System

Sep 2024 - Dec 2024

- Designed an IoT-based hydropower monitoring system integrating **6+ environmental and electrical sensors**
- Implemented embedded logic in Arduino for automated data collection and alert triggering
- Applied edge preprocessing to reduce cloud transmission load by filtering raw sensor noise
- Transmitted live sensor data using MQTT protocol for remote monitoring

Bash GUI System Tool

Dec 2023 - Feb 2024

- Developed GUI-based administrative tool using Bash and Zenity
- Automated secure file management operations, reducing manual configuration steps

Investment Trading Platform

Oct 2023 - Feb 2024

- Developed a secure authentication and portfolio management system using SQLite
- Implemented transaction validation and data integrity checks, reducing processing delays by **20%**

LEADERSHIP EXPERIENCE

Middlesex University	London, UK
Student Voice Leader	Nov 2023 - Present
- Represented over 100 students and facilitated communication to enhance digital resource delivery. - Chaired monthly meetings focused on improving digital security and academic support initiatives.	

EDUCATION

Middlesex University	London, UK
Computer networks and security	Graduation Date: Apr 2026
Gujarat Technological University	Vadodara
Computer Engineering	Graduation Date: Oct 2021

SKILLS & INTERESTS

Networking:	TCP/IP, Subnetting, VLANs, OSPF, BGP (conceptual), NAT, ACLs, VPN, OpenVPN, Cisco IOS, ASA Firewalls, Switch & Router Configuration, Wireshark, Network Troubleshooting
Cybersecurity:	Wireless Penetration Testing, WPA2/WPA3, Aircrack-ng, Hashcat, Vulnerability Assessment, Risk Analysis, IDS/IPS, Threat Modelling, Security Hardening
Cloud & DevOps	AWS (EC2, IAM, Lambda, DynamoDB, S3, CloudWatch, GuardDuty, CloudTrail), Docker, Kubernetes (EKS), Serverless Architecture
Programming & Systems	Python, Bash, C, SQL, Linux Administration, System Monitoring, Git, GitHub
Soft Skills	Team leadership, collaboration, presentations, troubleshooting, debugging, performance optimization, time management, task prioritization, project delivery, analytical problem solving
Interests:	Chess, Table Tennis, Meditation, Cooking, Cricket